Do neutron-star glitches keep time?

A pre-registered, measurement-aware test of the vortex-creep waiting-time prediction

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Abstract

The vortex-creep model of pulsar glitches predicts the time to the next glitch from the fractional steps in spin frequency, $\Delta\nu/\nu$, and spin-down rate, $\Delta\dot{\nu}/\dot{\nu}$, of the last one (Zhou et al. 2022, equation 48). I test this within pulsars, using 225 inter-glitch intervals (219 completed, 6 censored) in 21 pulsars from the Jodrell Bank Glitch Catalogue. Two obstacles arise: the predictor contains the glitch size, which may set waiting times by itself, and published $\Delta\dot{\nu}$ values come from fitting methods that can couple the measurement to the following interval. I therefore separate a scaling test from a test of the $\Delta\dot{\nu}$ coefficient at fixed size, audit how each $\Delta\dot{\nu}$ was measured, emulate every method on synthetic spin-down histories, and calibrate all statistics against simulated populations in which the prediction holds, is absent, is replaced by size alone, or is imitated by measurement bias. The rules were frozen, with externally recorded checksums, before any statistic linking glitch properties to observed waits was computed, and the analysis was run once. Waiting times scale with the predicted ratio (slope 0.27, 95 per cent interval 0.20–0.34), and the spin-down step carries the predicted negative coefficient at fixed size (-0.26; -0.38 to -0.13), which neither size-only nor bias-only populations reproduce. The unit slope is not recovered, consistent with attenuation by measurement, and post hoc analyses show that the spin-down dependence varies between pulsars. An unphysical inertia ratio inferred for the Crab pulsar is traced to catalogue $\Delta\dot{\nu}$ values that include fast transients, which equation 48 excludes.

Key words: stars: neutron – pulsars: general – dense matter – methods: statistical

1 Introduction

A pulsar glitch is a sudden increase in rotation frequency, usually accompanied by an increase in the magnitude of the spin-down rate that then relaxes over days to years (Espinoza et al., 2011; Yu et al., 2013; Basu et al., 2022). More than seven hundred glitches are now catalogued in over two hundred pulsars. The favoured explanation places the angular-momentum reservoir in a neutron superfluid whose quantised vortices are pinned to the inner-crust lattice (Anderson & Itoh, 1975); when pinning fails, vortices move outward and transfer angular momentum to the crust (for reviews see Haskell & Melatos, 2015; Antonopoulou, Haskell & Espinoza, 2022; Zhou et al., 2022).

The vortex-creep model (Alpar et al., 1984) adds a mechanism for the inter-glitch evolution. Pinned vortices migrate outward by thermally activated creep, so the lag between superfluid and crust settles to a steady value. A glitch empties some creep regions and reduces the lag; spin-down then rebuilds it. If the next glitch occurs once the lag has been restored, each glitch should foretell the interval that follows it. Alpar & Baykal (2006) wrote the relations between the observed steps and the moments of inertia of the regions involved, and Zhou et al. (2022, their equation 48) combined them into an explicit waiting-time prediction in terms of $\Delta\nu/\nu$ and $\Delta\dot{\nu}/\dot{\nu}$ (Section 2).

Earlier confrontations of creep timing with data have worked glitch by glitch, using the inter-glitch second derivative of the frequency obtained from timing fits. Akbal et al. (2017) estimated the time to the next glitch for 16 large glitches of the Vela pulsar from the step in spin-down rate and the fitted inter-glitch $\ddot{\nu}$. They found the estimates broadly consistent with the observed intervals once persistent shifts in the spin-down rate were allowed for. Güger-cinoğlu & Alpar (2020) applied the same framework to the 2016 Vela glitch, and Akbal, Gügercinoğlu & Alpar (2021) to 45 glitches of PSR J0537−6910. These analyses require a timing solution for every inter-glitch interval. They are also, by construction, single-pulsar studies.

This paper asks a narrower question with a broader sample: does the relation, evaluated only with catalogued quantities, predict waiting times within pulsars across the glitching population? Two difficulties stand in the way.

Glitch size can do the work alone. In PSR J0537−6910 the interval to the next glitch grows almost linearly with the size of the preceding glitch (Middleditch et al., 2006). The rank correlation reaches 0.89 (Zubieta et al., 2026). Any predictor that contains $\Delta\nu$ will therefore correlate with waiting times whatever the mechanism. Only the dependence on $\Delta\dot{\nu}$ at fixed size is specific to the creep relation.

$\Delta\dot{\nu}$ is not one observable. Catalogue values have been obtained by differencing short timing solutions on either side of the glitch (Espinoza et al., 2011), as a persistent step in fits bounded by neighbouring glitches (Antonopoulou et al., 2018), and as the instantaneous total, including exponential terms, in simultaneous fits to all glitches of a pulsar (Basu et al., 2022). A step averaged over the post-glitch interval is mechanically correlated with that interval's length, so a careless choice of measurement could create or erase the very association under test.

The analysis here is designed around both difficulties:

1. the prediction is split into a scaling test and a test of the $\Delta\dot{\nu}$ coefficient at fixed glitch size (Section 2);
2. every included $\Delta\dot{\nu}$ is traced to its source paper and measurement method, each method is emulated on synthetic spin-down series, and every decision statistic is referred to simulated populations passed through

the same emulation (Sections 3–5);

3. all rules were frozen before any statistic relating glitch properties to observed waiting times was computed, and the analysis was run once (Section 5.4).

The design is validated on simulations in Section 6. The pre-registered result is given in Section 7, analyses made afterwards are reported separately in Section 8, and the interpretation and limitations are discussed in Section 9.

2 The prediction

Following Alpar & Baykal (2006) and Zhou et al. (2022), let $I_{\rm A}$ be the moment of inertia of the non-linear creep regions through which vortices move at a glitch, $I_{\rm B}$ that of the regions they cross without creeping, and I the total. Assuming a linear decrease of the superfluid angular velocity across $I_{\rm A}$, and once the prompt exponential relaxations are over,

$$\frac{\Delta\nu}{\nu} = \left(\frac{I_{\rm A}}{2I} + \frac{I_{\rm B}}{I}\right)\frac{\delta\Omega_{\rm s}}{\Omega}, \qquad \frac{\Delta\dot{\nu}}{\dot{\nu}} = \frac{I_{\rm A}}{I}, \tag{1}$$

where $\delta\Omega_{\rm s}$ is the glitch-induced decrease of the superfluid angular velocity (Zhou et al., 2022, equations 45 and 46). The lag is restored after $t_0 = \delta\Omega_{\rm s}/|\dot{\Omega}|$, which gives

$$t_0 = 2\times10^{-3}\,\frac{(\Delta\nu/\nu)_{-6}}{(\beta+\frac{1}{2})\,(\Delta\dot{\nu}/\dot{\nu})_{-3}}\,\tau_{\rm sd}, \tag{2}$$

with $\beta \equiv I_{\rm B}/I_{\rm A}$ and $\tau_{\rm sd} \equiv \nu/(2|\dot{\nu}|)$ (Zhou et al., 2022, equation 48). Subscripts -6 and -3 denote units of 10^{-6} and 10^{-3}, and t_0 is in the units of $\tau_{\rm sd}$.

Writing $R \equiv (\Delta\nu/\nu)_{-6}/(\Delta\dot{\nu}/\dot{\nu})_{-3}$, identifying t_0 with the observed wait T and allowing log-normal scatter gives, for glitch k of pulsar p,

$$\ln T_{pk} = \ln R_{pk} + c_p + \sigma\varepsilon_{pk}, \qquad {\rm e}^{c_p} = \frac{2\times10^{-3}\,\tau_{{\rm sd},p}}{\beta_p+\frac{1}{2}}, \tag{3}$$

with $\varepsilon_{pk} \sim \mathcal{N}(0,1)$. Three features of equation (3) shape the test.

First, $\ln(\Delta\nu/\nu)$ enters with coefficient $+1$ and $\ln(\Delta\dot{\nu}/\dot{\nu})$ with -1. The second coefficient is the part of the prediction that a dependence of waiting time on glitch size alone cannot reproduce.

Secondly, β_p is not predicted and is absorbed in a per-pulsar intercept. The test is therefore within pulsars, and assumes that β_p is constant from glitch to glitch in a given pulsar. That assumption is needed to write equation (3); it is not part of the theory as stated. Inverting the intercept gives an implied $\beta_p = 2\times10^{-3}\tau_{{\rm sd},p}{\rm e}^{-c_p} - \frac{1}{2}$, which must be non-negative.

Thirdly, equation (1) holds after the exponential transients have decayed. The $\Delta\dot{\nu}$ of equation (2) is therefore the step that persists, not an instantaneous total that includes the transient terms. Section 9.2 returns to this point.

3 Data

3.1 Catalogue and intervals

Glitch epochs, $\Delta\nu/\nu$, $\Delta\dot{\nu}/\dot{\nu}$ and their uncertainties were taken from the Jodrell Bank Glitch Catalogue[1] (Espinoza et al., 2011; Basu et al., 2022), downloaded on 2026 October 30 (736 glitches in 226 objects; SHA-256 of the file in Section 5.4). Spin frequencies and derivatives for $\tau_{\rm sd}$ are from the ATNF Pulsar Catalogue, version 2.9.0 (Manchester et al., 2005).

Eight objects on a pre-specified list, catalogued under X-ray designations and mostly magnetars, were removed, leaving 715 glitches in 218 pulsars. A glitch is *usable* if both $\Delta\nu > 0$ and $\Delta\dot{\nu} > 0$ are catalogued (503 glitches). Each usable glitch starts an interval that ends at the next glitch of the same pulsar, whether or not that glitch is usable; glitches are ordered by epoch, not by catalogue number. A pulsar's final interval is right-censored at a documented end of monitoring where one exists (e.g. Basu et al., 2022; Gügercinoğlu et al., 2022; Ho et al., 2026) and is otherwise dropped (138 final intervals). The gap between RXTE and NICER monitoring of PSR J0537−6910 (Antonopoulou et al., 2018; Ho et al., 2026) censors the one interval that spans it. No glitch is ever removed for epoch uncertainty, because removing one would merge two intervals into a spuriously long one.

3.2 Source audit

For every source paper behind a glitch in the candidate pulsars, I recorded how $\Delta\dot{\nu}$ was obtained (Appendix B). Each glitch was assigned to one of five classes:

- *local relaxation*: a step fitted to data bounded by the neighbouring glitches, with recoveries modelled (Antonopoulou et al., 2018; Yu et al., 2013; Ho et al., 2026);
- *global with exponentials*: all glitches and detected recoveries of a pulsar fitted simultaneously (Basu et al., 2022; Lower et al., 2021; Gügercinoğlu et al., 2022);
- *short window*: the difference between timing solutions over analyst-chosen spans on either side of the glitch, transients not separated (Espinoza et al., 2011; Wang et al., 2000);
- *window*: a level averaged up to the next glitch;
- *global without exponentials*, or *unknown*.

The first three are admissible. Short-window values were admitted only because the emulation of Section 4 showed, under a rule fixed in advance, that their bias on the forward slope stays below a set limit (largest upper 95 per cent bound 0.016 against a limit of 0.02; a rerun after the recovery settings were updated, Section 4, gives 0.013). Window-type measurements were excluded because the emulation shows that they can imitate the predicted signal (Section 6). Glitches from sources that could not be audited are classed as unknown and excluded. The method used for the catalogue maintainers' own unpublished measurements is assumed to follow Basu et al. (2022); a robustness case removes them.

[1] `https://www.jb.man.ac.uk/pulsar/glitches/gTable.html`

3.3 Sample

A pulsar enters if it has at least four admissible completed intervals. Twenty-one pulsars qualify, with 225 admissible intervals: 219 completed and 6 censored (Table 1). Of these, 70 are measured by local relaxation, 55 by global fits with exponentials, and 100 by short windows. The sample is uneven. PSR J0537−6910 contributes 63 intervals (28 per cent), PSR B1737−30 28 and the Crab pulsar 24, while fifteen pulsars contribute four to six each. The Vela pulsar does not enter: its 22 usable glitches come from 15 sources with incompatible definitions of $\Delta\dot{\nu}$ (Section 9.3).

4 Emulating the measurement of $\Delta\dot{\nu}$

The purpose of the emulation is to learn, for the actual mix of methods, cadences and uncertainties in the sample, what each fitting method returns when the truth is known.

4.1 Synthetic spin-down histories

For each pulsar, synthetic glitch sequences are generated across its real observing window. The properties of each synthetic glitch ($\Delta\nu/\nu$, $\Delta\dot{\nu}/\dot{\nu}$, their uncertainties and the source's cadence) are resampled from that pulsar's admissible glitches. Between glitches, the magnitude of the spin-down rate is

$$|\dot{\nu}|(t) = |\dot{\nu}_0| + \sum_{k:t_k<t} \left[J_k \left(1 - \frac{t-t_k}{t_{\mathrm{rec},k}}\right)_+ + \frac{Q_k \Delta\nu_k}{\tau_{\mathrm{d},k}} \, \mathrm{e}^{-(t-t_k)/\tau_{\mathrm{d},k}} \right], \tag{4}$$

where J_k is the true step, $(x)_+ = \max(x, 0)$, and the second term is the spin-down equivalent of an exponential recovery of fractional amplitude Q_k and time-scale $\tau_{\mathrm{d},k}$. The linear relaxation of J_k over t_{rec} follows the creep picture: in model-true simulations $t_{\mathrm{rec}} = T$, so the step has relaxed exactly when the next glitch occurs. The series is sampled at the source's cadence. Gaussian noise is added, with its level set for each pulsar by bisection so that the median formal relative uncertainty of the emulated $\Delta\dot{\nu}$ matches the median catalogue $\sigma_{\Delta\dot{\nu}}/\Delta\dot{\nu}$ of that pulsar.

The recovery parameters are drawn log-uniformly within the envelope of published single-component fits:

- PSR J0537−6910: $Q \in [0.002, 0.04]$, $\tau_{\mathrm{d}} \in [4, 37]$ d, applied only to glitches with $\Delta\nu \geq 20\,\mu$Hz, the only ones with detected recoveries (Zubieta et al., 2026);
- the Crab pulsar: $Q \in [0.58, 1.00]$, $\tau_{\mathrm{d}} \in [2.0, 21.1]$ d (Wong, Backer & Lyne, 2001; Wang et al., 2012);
- all other pulsars, from Lower et al. (2021): for $\Delta\nu/\nu \geq 10^{-7}$, $Q \in [0.002, 0.06]$ and $\tau_{\mathrm{d}} \in [5.8, 488]$ d; for smaller glitches, $Q \in [0.10, 0.15]$ and $\tau_{\mathrm{d}} \in [54, 2340]$ d.

4.2 Measurement

$\Delta\dot{\nu}$ is then measured from the synthetic series as each method class would measure it. In the local-relaxation and global classes, an exponential term enters the fit only if it lowers the Akaike information criterion by at least 10. A global fit includes every glitch and every retained exponential of the pulsar at once, with one $\ddot{\nu}$. A short-window measurement differences constant levels fitted over $w = 40$ d (the post-glitch span of Espinoza et al., 2011) or $w = 125$ d (Wang et al., 2000) on either side of the glitch, truncated at neighbouring glitches, without exponential terms. A window measurement differences the level averaged up to the next glitch from the pre-glitch level. A glitch with fewer than four post-glitch samples has no measurable $\Delta\dot{\nu}$: it ends the previous interval but starts none, as in the real data.

Unit tests check the emulator against analytic cases. For example, a window measurement of a linearly relaxing step returns $J(1 - T/2t_{\mathrm{rec}})$ for $T \leq t_{\mathrm{rec}}$, and global and local fits with every exponential retained recover the injected values exactly.

4.3 Limits of the emulation

The emulation works in the $\dot{\nu}$ domain rather than fitting phase-connected times of arrival, and contains no red timing noise. It omits the slow growth of the Crab pulsar's persistent spin-down step, about 46 per cent of which develops on a 320-d time-scale (Lyne et al., 2015), as well as multi-component recoveries. Two settings are not taken from tabulated values: the scatter of t_{rec} in the bias-only simulations (log-normal, standard deviation 0.3 about the pulsar's typical wait) and the short-window spans. The emulation also detects recoveries in PSR J0537−6910 more often than the published analysis does (in 19–27 per cent of glitches, against 12 of 66; Zubieta et al. 2026). For four to six of the 21 pulsars, depending on the generating σ (four at $\sigma = 0.2$, five at 0.4–0.8 and six at 1.2; the Crab pulsar at every σ), the emulated $\Delta\dot{\nu}$ uncertainties exceed the catalogue values even with no added noise, so their simulated measurements are noisier than the real ones.

5 Statistical framework

5.1 Models

Waiting times are modelled with log-normal accelerated-failure-time regressions. Each has one intercept per pulsar and a scatter σ shared by all pulsars. Completed intervals contribute their density and censored ones their survival probability:

$$\ln L = \sum_{\mathrm{compl.}} \left[\ln\phi\left(\frac{\ln T - \mu}{\sigma}\right) - \ln\sigma - \ln T \right] + \sum_{\mathrm{cens.}} \ln\left[1 - \Phi\left(\frac{\ln c - \mu}{\sigma}\right)\right], \tag{5}$$

where ϕ and Φ are the standard normal density and distribution function and c is the censoring time. The linear predictor μ defines five models:

$$\begin{aligned}
&\mathrm{M0}: \mu = c_p, \\
&\mathrm{M1}: \mu = \ln R + c_p \quad \text{(equation 3 literally)}, \\
&\mathrm{M2}: \mu = b \ln R + c_p, \\
&\mathrm{M3}: \mu = a \ln(\Delta\nu/\nu) + c_p, \\
&\mathrm{M4}: \mu = a \ln(\Delta\nu/\nu) + d \ln(\Delta\dot{\nu}/\dot{\nu}) + c_p.
\end{aligned}$$

The prediction is $b > 0$ in M2 and $d < 0$ in M4, with $b = 1$ and $d = -1$ in the absence of measurement error. Fits

Table 1: The 21 pulsars of the primary sample, ordered by number of admissible intervals. ν and $\dot{\nu}$ are from the ATNF catalogue (Manchester et al., 2005); $\tau_{\rm sd} = \nu/(2|\dot{\nu}|)$. Intervals are admissible intervals, with the number censored in parentheses, followed by their $\Delta\dot{\nu}$ method class: LR, local relaxation; G, global with exponentials; SW, short window.

Pulsar	J name	ν (Hz)	$\dot{\nu}$ (10^{-12} Hz s^{-1})	$\tau_{\rm sd}$ (kyr)	Intervals	LR	G	SW
PSR J0537−6910	J0537−6910	62.026	−199	4.93	63 (2)	63	0	0
PSR B1737−30	J1740−3015	1.648	−1.27	20.6	28 (0)	0	4	24
PSR B0531+21	J0534+2200	29.947	−378	1.26	24 (0)	0	1	23
PSR B1338−62	J1341−6220	5.172	−6.77	12.1	16 (0)	1	7	8
PSR J0631+1036	J0631+1036	3.474	−1.24	44.4	12 (1)	0	3	9
PSR J2229+6114	J2229+6114	19.362	−29	10.6	8 (0)	0	6	2
PSR J1841−0524	J1841−0524	2.243	−1.17	30.3	6 (0)	0	3	3
PSR J0205+6449	J0205+6449	15.217	−44.9	5.37	6 (1)	0	6	0
PSR B1757−24	J1801−2451	8.005	−8.2	15.5	6 (0)	0	1	5
PSR J1023−5746	J1023−5746	8.971	−30.9	4.6	6 (1)	0	6	0
PSR B1758−23	J1801−2304	2.405	−0.653	58.3	6 (0)	0	2	4
PSR J1814−1744	J1814−1744	0.252	−0.0471	84.6	5 (1)	0	2	3
PSR B1046−58	J1048−5832	8.081	−6.28	20.4	5 (0)	2	1	2
PSR B1800−21	J1803−2137	7.481	−7.52	15.8	5 (0)	0	1	4
PSR J1420−6048	J1420−6048	14.667	−17.9	13	5 (0)	3	2	0
PSR B1727−47	J1731−4744	1.205	−0.238	80.4	4 (0)	1	0	3
PSR B1823−13	J1826−1334	9.853	−7.31	21.4	4 (0)	0	1	3
PSR J0729−1448	J0729−1448	3.973	−1.79	35.2	4 (0)	0	0	4
PSR J1617−5055	J1617−5055	14.418	−28.1	8.13	4 (0)	0	4	0
PSR J1737−3137	J1737−3137	2.220	−0.684	51.4	4 (0)	0	2	2
PSR J2021+3651	J2021+3651	9.639	−8.89	17.2	4 (0)	0	3	1
Total					225 (6)	70	55	100

use Newton iterations with step halving, and the M0 likelihood is cross-checked against an independent implementation (LIFELINES; Davidson-Pilon, 2019).

5.2 Simulated populations

Every decision refers a real statistic to its distribution over 1000 simulated catalogues (500 for bias-only populations). These share the real pulsars, glitch counts, observing windows, method mix and uncertainties:

- **S1** – equation (3) is true, with catalogue-level measurement noise only;
- **S1m** – as S1, but $\Delta\dot{\nu}$ is measured through the emulation with $t_{\rm rec} = T$;
- **S2** – waits independent of glitch properties;
- **S3, S3m** – waits depend on glitch size alone, $\ln T = a\ln(\Delta\nu/\nu) + c_p + \sigma\varepsilon$, without and with the emulation;
- **S4** variants – waits as in S2, but $\Delta\dot{\nu}$ measured through each method class with $t_{\rm rec}$ independent of T, so that any association is pure measurement artefact; **S4-mix** uses the real method mix.

Covariates are centred within pulsars. Intercepts are set so that the mean synthetic wait equals the pulsar's observed glitch rate, which is the only information from real waiting times used in simulation.

The fitted scatter $\hat{\sigma}$ exceeds the generating σ when measurement attenuates the slope, because the lost dependence ends up in the residuals. Each population is therefore simulated at the generating σ that reproduces the real $\hat{\sigma}$ of M2, found by bisection on $[0.01, 2]$ (200 replicates per step, tolerance 0.01). A population that cannot reach the real $\hat{\sigma}$ even at $\sigma = 0.01$ is run at that lower bound and the fact is reported.

5.3 Tests and outcomes

T1 (scaling). The signed root of the likelihood-ratio statistic for M2 against M0, positive when $\hat{b} > 0$, must exceed its 97.5th percentile under S2.

T2 (the $\Delta\dot{\nu}$ term). $\hat{d}$ in M4 must fall below the smaller of its 2.5th percentiles under S3 and S3m. The size coefficient of those populations is set to the real M3 estimate.

Both tests are one-sided in the predicted direction at level 0.025, which gives a family-wise level of 0.05. The outcome is the first of the following to apply:

D *artefact signature*: the placebo slope, from M2 with the previous wait T_{k-1} as outcome, lies below its 2.5th percentile under S1m;

C *refutation*: $\hat{b}$ lies below its 2.5th percentile under S1m;

A *scaling supported*: T1 rejects, and $\hat{b}$ exceeds the 97.5th percentile under S4-mix, is at least the 2.5th percentile under S1m, and is at most the 97.5th percentile under S1. Within A:

- **A2** if T2 rejects (creep-specific support);
- **B** if T2 does not reject although its power is at least 0.8 ($\Delta\dot{\nu}$ term ruled out);
- **A1** otherwise (an association that a size-only population also produces);

D *signal within bias range*: T1 rejects but $\hat{b}$ does not exceed the S4-mix 97.5th percentile;

Inconclusive: none of the above.

Reported intervals for b and d are Wald intervals widened by the ratio of the simulated spread of the estimate to its mean model standard error under the matching null (S2 for b, S3m for d). Secondary analyses are reported without decisions: per-pulsar Spearman correlations pooled by Fisher's z (completed intervals only), a within-pulsar permutation test of $\hat{b}$ (10^4 permutations), leave-one-out predictive scores, a conditional logistic regression of missing $\Delta\dot{\nu}$ on the following wait, and the implied β_p.

5.4 Pre-registration

The decision rules (D1–D15) were written on 2026 October 31 and amended sixteen times (D4a–D4p) on 2026 October 31 and November 1 as validation on simulated data exposed problems. Every amendment and its reason is listed in Appendix A. Several amendments were prompted by simulated power or error rates; none was made after any statistic linking glitch properties to real waiting times had been computed. A guard in the code refused any such statistic until a freeze file existed and the analysis data set matched a recorded checksum.

The rules were frozen on 2026 November 1, and the following SHA-256 digests were sent to an external e-mail account the same day:

- rules (`DECISIONS.md`): `2fd950fe...0bed`;
- configuration: `3cd2b2f9...671f`;
- analysis data set: `27a63d37...be3b`;
- catalogue file: `5ba59d01...49de7`.

Between the freeze and the primary run, one commit replaced a unit test that asserted the project was not yet frozen; no analysis code, rule or data changed. Code added later serves only the post hoc analyses, the forecast and reproduction, and changes no pre-registered result. The primary script refuses to run once its output exists, and it was run once.

6 Validation

All 105 pre-specified validation criteria were met, with no failed replicate in any simulation job. The criteria were evaluated at five generating values $\sigma \in \{0.2, 0.4, 0.6, 0.8, 1.2\}$, with 1000 replicates per population (500 for S4 variants) and seed 20261031. False-positive rates were required not to exceed $0.025 + 3\sqrt{0.025 \times 0.975/n} \approx 0.040$. Observed rates were:

- T1 under S2: 0.017–0.031;
- T2 under S2: 0.001–0.012, and under S3: 0.004–0.020;
- outcome A2 under the measured size-only population S3m: 0.016–0.024;

Table 2: Power of the frozen design under the model-true population S1m, and the creep-specific false-positive rate under the measured size-only population S3m, as functions of the generating σ. The second row gives the corresponding mean fitted $\hat{\sigma}$ of M2.

Generating σ	0.2	0.4	0.6	0.8	1.2
Mean $\hat{\sigma}$ (S1m)	0.76	0.80	0.87	0.94	1.09
T1 power	0.990	0.980	0.952	0.902	0.492
T2 power	0.954	0.828	0.597	0.371	0.109
A2 rate, S1m	0.916	0.799	0.581	0.357	0.104
A2 rate, S3m	0.016	0.021	0.024	0.020	0.023
S1m $\hat{b}$, 2.5%	0.22	0.20	0.13	0.09	−0.03
S1m $\hat{b}$, 97.5%	0.67	0.62	0.59	0.57	0.47

- outcome A2 under the bias-only mix S4-mix: at most 0.010.

With measurement noise switched off, the median $\hat{b}$ under S1 lay within 0.035 of 1 at every σ.

Table 2 gives the power of the frozen design. Figure 1 shows the T1 rejection rate and mean slope for every population.

Three results of the validation shape the interpretation.

1. *Attenuation.* Even when equation (3) holds exactly, realistic measurement of $\Delta\dot{\nu}$ pulls the expected slope from 1 down to roughly 0.1–0.7 (Table 2). Support is therefore judged against the emulated range, not against unity.
2. *A residual floor.* Under S1m the fitted $\hat{\sigma}$ cannot fall below about 0.74 (95 per cent range 0.61–0.85 at $\sigma = 0.01$). This floor is set by the measurement-induced loss of signal, not by intrinsic scatter.
3. *Window bias.* Window-type measurement alone produces outcome A2 in up to 60 per cent of replicates at large σ, which justifies its exclusion. The other single-method bias populations rarely do so (A2 rate at most 0.10, and at most 0.010 for S4-mix).

7 Pre-registered result

The primary fit gives $\hat{\sigma} = 0.696$ for M2, compared with 0.787 for M0. Table 3 lists the quantities that decide the outcome; Fig. 2 shows the data pulsar by pulsar.

Applying the rules in order: the placebo slope is not below its bound, so outcome D does not apply. $\hat{b}$ is not below the S1m 2.5th percentile, so C does not apply. T1 rejects and $\hat{b}$ satisfies all three bounds, so A applies. Because T2 rejects, the outcome is **A2**.

The estimate $\hat{b} = 0.268$ lies at the 4.5th percentile of the model-true population S1m, inside but near its lower edge. The bias-only population reaches only 0.111. The secondary analyses agree with the primary one:

- the within-pulsar permutation test of $\hat{b}$ gives $p = 1.0 \times 10^{-4}$ (the smallest value 10^4 permutations allow);
- the pooled Spearman correlation between $\ln R$ and $\ln T$ is 0.60 (0.49–0.69; 21 pulsars, completed intervals only);

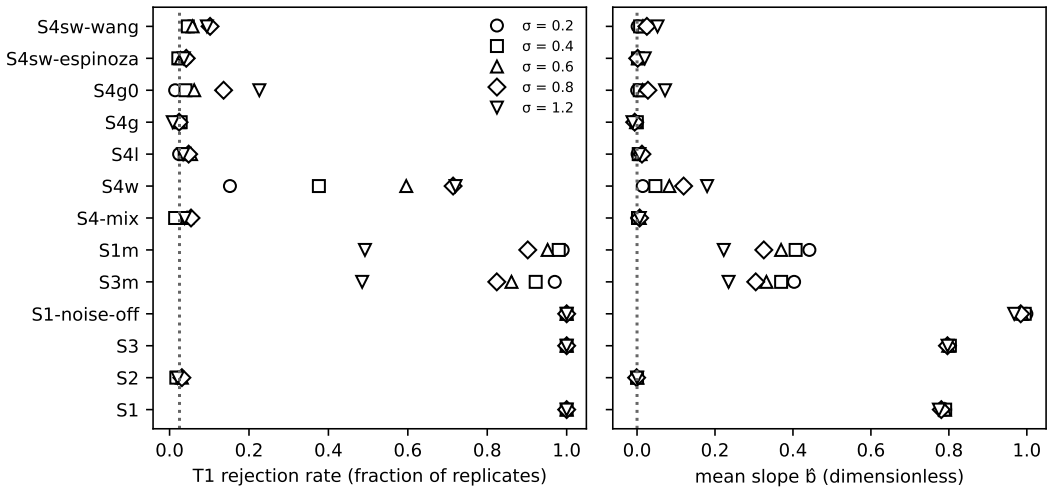


Figure 1: Validation of the frozen design. Left: fraction of replicates in which T1 rejects; right: mean $\hat{b}$. Each row is a simulated population (Section 5.2) and each symbol a generating σ. The dotted line marks the nominal level 0.025 (left) and zero slope (right).

Table 3: Pre-registered quantities and the simulated references that decide the outcome. Intervals are 95 per cent; calibrated intervals widen the Wald interval by the simulated-spread factor of the matching null (1.015 for b, 1.050 for d). Percentiles of reference populations are evaluated at the matched generating σ of each population.

Quantity	Estimate	Interval	Reference	Verdict
Placebo slope (M2 on T_{k-1}, $n = 211$)	-0.026	$[-0.108, 0.056]$	-0.357 (S1m 2.5th)	not below: no artefact signature
$\hat{b}$ vs refutation bound	0.268	$[0.200, 0.336]$	0.230 (S1m 2.5th)	not below: no refutation
T1 signed root (M2 vs M0)	7.398	–	2.037 (S2 97.5th)	rejects
$\hat{b}$ vs bias-only	0.268		0.111 (S4-mix 97.5th)	exceeds
$\hat{b}$ vs model ceiling	0.268		0.926 (S1 97.5th)	below
$\hat{d}$ (T2, M4 vs M3)	-0.257	$[-0.384, -0.130]$	-0.130 (S3m 2.5th; S3: -0.094)	below: rejects

Outcome: A2, scaling supported with the $\Delta\dot{v}$ term established. Power at the matched σ: T1 0.997, T2 0.769.

Table 4: Implied $\beta_p = 2 \times 10^{-3} \tau_{\mathrm{sd},p} \mathrm{e}^{-c_p} - \frac{1}{2}$ with 95 per cent intervals, from the intercepts of M1. Class is the pre-registered D7 split (Vela-like if the median usable $\Delta\nu/\nu \geq 10^{-6}$).

Pulsar	n	Class	c_p	β	95% interval
PSR J0537−6910	63	Crab-like	5.50	14.22	[9.74, 18.70]
PSR B1737−30	28	Crab-like	9.29	0.90	[0.26, 1.53]
PSR B0531+21	24	Crab-like	10.24	−0.47	[−0.48, −0.45]
PSR B1338−62	16	Crab-like	7.28	5.59	[1.94, 9.25]
PSR J0631+1036	12	Crab-like	9.59	1.72	[0.13, 3.31]
PSR J2229+6114	8	Crab-like	9.59	0.03	[−0.42, 0.48]
PSR B1757−24	6	Vela-like	8.79	1.22	[−0.47, 2.90]
PSR B1758−23	6	Crab-like	9.65	2.26	[−0.45, 4.96]
PSR J0205+6449	6	Vela-like	8.61	0.21	[−0.53, 0.96]
PSR J1023−5746	6	Vela-like	7.77	0.92	[−0.52, 2.37]
PSR J1841−0524	6	Crab-like	8.09	6.26	[−0.37, 12.89]
PSR B1046−58	5	Vela-like	8.01	4.44	[−0.86, 9.74]
PSR B1800−21	5	Vela-like	9.18	0.68	[−0.59, 1.96]
PSR J1420−6048	5	Vela-like	8.28	1.91	[−0.68, 4.49]
PSR J1814−1744	5	Crab-like	12.30	−0.22	[−0.54, 0.10]
PSR B1727−47	4	Crab-like	11.65	0.01	[−0.60, 0.62]
PSR B1823−13	4	Vela-like	8.67	2.18	[−1.04, 5.40]
PSR J0729−1448	4	Crab-like	10.37	0.31	[−0.66, 1.27]
PSR J1617−5055	4	Crab-like	9.35	0.02	[−0.60, 0.63]
PSR J1737−3137	4	Crab-like	8.92	4.53	[−1.51, 10.57]
PSR J2021+3651	4	Vela-like	8.77	1.46	[−0.89, 3.80]

- the corresponding placebo correlation is -0.09 (-0.25 to 0.07), with permutation $p = 0.55$.

Leave-one-out predictive scores (mean log score per interval) rank the models M2 (-6.834), M4 (-6.848), M3 (-6.859), M0 (-6.938) and M1 (-7.393; Fig. 3). The literal relation with unit slope predicts worse than intercepts alone.

Two diagnostics complete the pre-registered set.

Missing $\Delta\dot{\nu}$. Glitches without a catalogued $\Delta\dot{\nu}$ tend to be followed by shorter waits (odds ratio 0.55, 0.33–0.92, $p = 0.022$). Such an association is expected when a short wait leaves too little data to measure the step. The observed log odds ratio (-0.59) lies inside the range produced by S1m (-1.40 to -0.59), at its 97th percentile.

Implied β (Table 4; Fig. 4). The implied ratio is positive or consistent with zero for 20 of the 21 pulsars. For the Crab pulsar it is -0.47 (-0.48 to -0.45), which is unphysical; Section 9.2 traces the cause. PSR J0537−6910 gives $\beta = 14.2$ (9.7–18.7).

7.1 Robustness

The pre-registered robustness cases (Table 5) receive no outcome. Their intervals reuse the primary calibration factors and are approximate. The slope $\hat{b}$ keeps its sign in every case. $\hat{d}$ keeps its sign in every case except four: the global-fit class (33 intervals), PSR B1338−62 alone (16), PSR J0631+1036 alone (12) and the Vela-like group (41). All four intervals are wide and include zero. The test of step-only $\Delta\dot{\nu}$ values could not be run, because the catalogue does not record them.

8 Post hoc analyses

The analyses in this section were chosen after the primary result was known. They are not part of the pre-registration, cannot change the outcome, and are reported to show where the result comes from.

Ratio form. This comparison uses only the pre-registered fits of M2 and M4, but was not itself specified in advance. M4 and M2 reach almost the same likelihood ($\Delta \ln L = 0.026$), with $\hat{a} = 0.268$ and $\hat{d} = -0.257$ in M4. The data are thus consistent with $\Delta\nu/\nu$ and $\Delta\dot{\nu}/\dot{\nu}$ entering only through their ratio, as equation (2) requires. Because measurement is expected to attenuate d more than a, this near-equality should not be over-read.

Intervals shorter than the fit window. If the next glitch falls inside a short-window fit span, the measured step and the following wait are coupled. Four of the 100 short-window intervals are shorter than the span. Without them $\hat{d} = -0.179$ (-0.304 to -0.053).

Influence. Removing any single interval leaves $\hat{d}$ between -0.278 and -0.208 (225 refits, none failed). Of the five most influential intervals, three are short waits of PSR B1737−30 (26, 34 and 48 d; ranks 1, 2 and 5); the other two are long waits of PSR B1758−23 (1154 d) and PSR J2229+6114 (352.5 d).

Single-pulsar references. For the three pulsars with the most intervals, each pulsar was simulated alone under S3m, under its own-method bias-only population and under S1m, with σ matched to its own M4 fit (Table 6).

- In PSR B1737−30 and PSR J0537−6910, which are measured by different methods, $\hat{d}$ lies beyond the size-only and bias-only ranges and within the model-true range.
- For the Crab pulsar all three ranges are so wide that the test is uninformative.

Heterogeneity. For the 11 pulsars with at least six intervals (181 intervals), M4 with a separate size coefficient per pulsar was compared with and without a separate d per pulsar. The likelihood ratio is 28.2 for 10 degrees of freedom ($p = 0.0017$), with common $\hat{d} = -0.41 \pm 0.07$.

- PSR B1737−30 (-0.73 ± 0.11) and the Crab pulsar (-0.74 ± 0.18) show a strong effect.
- PSR J0537−6910 (-0.18 ± 0.15) shows a weak one.
- PSR B1338−62 ($+0.01 \pm 0.19$) and PSR J0631+1036 ($+0.13 \pm 0.23$) show none.

The nominal p-value overstates the evidence for two reasons. The model forces a common σ on pulsars whose single-pulsar fits range from $\hat{\sigma} = 0.30$ to 1.13. And different method mixes are expected to attenuate the measured d by different amounts even if the true d were universal.

9 Discussion

9.1 What the result does and does not show

Under rules fixed in advance, within-pulsar waiting times scale with the creep ratio R. The spin-down step carries the predicted negative sign at fixed glitch size, beyond what glitch size alone or the emulated measurement biases produce. That is the creep-specific content of equation (2), and the outcome is the strongest the design allowed.

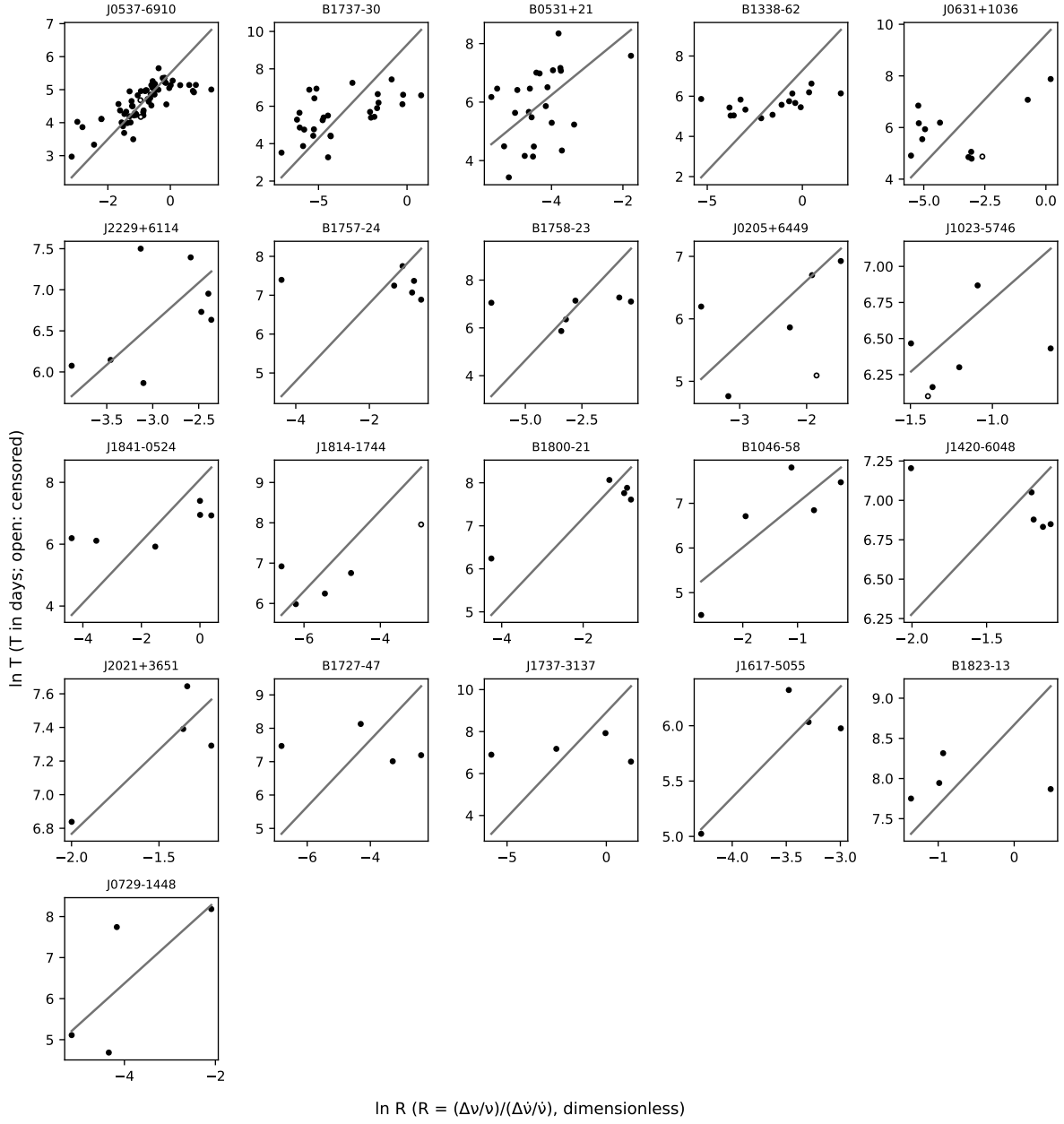


Figure 2: Waiting time against $R = (\Delta\nu/\nu)_{-6}/(\Delta\dot{\nu}/\dot{\nu})_{-3}$ for the 21 pulsars, ordered by number of intervals. Filled symbols are completed and open symbols censored intervals. Lines have unit slope, the literal prediction of equation (3); the fitted common slope is $\hat{b} = 0.27$.

Table 5: Pre-registered robustness cases (estimates reported, no outcome assigned). Intervals are approximate: they use the primary calibration factors.

Case	n	Pulsars	$\hat{b}$ [95%]	$\hat{d}$ [95%]	$\hat{\sigma}$
Primary	225	21	0.268 [0.200, 0.336]	−0.257 [−0.384, −0.130]	0.696
Without PSR J0537−6910	162	20	0.242 [0.161, 0.323]	−0.248 [−0.401, −0.096]	0.780
Without the Crab pulsar	201	20	0.255 [0.194, 0.316]	−0.208 [−0.327, −0.089]	0.613
Censored intervals dropped	219	21	0.262 [0.194, 0.330]	−0.240 [−0.369, −0.111]	0.695
$\sigma_{\ln R} \leq 0.5$	156	15	0.304 [0.226, 0.382]	−0.407 [−0.574, −0.240]	0.626
Gap-flagged intervals excluded	224	21	0.268 [0.201, 0.336]	−0.258 [−0.385, −0.131]	0.696
$\Delta\nu/\nu \geq 10^{-7}$	122	12	0.418 [0.325, 0.512]	−0.260 [−0.381, −0.138]	0.402
Epoch uncertainty $\leq 0.1\,T$	196	21	0.242 [0.168, 0.316]	−0.235 [−0.371, −0.100]	0.702
Later censoring for Jodrell Bank pulsars	231	21	0.283 [0.214, 0.353]	−0.271 [−0.400, −0.142]	0.717
Without maintainers' own measurements	208	18	0.280 [0.208, 0.352]	−0.266 [−0.399, −0.133]	0.702
Class: local relaxation	63	1	0.480 [0.378, 0.582]	−0.182 [−0.340, −0.024]	0.351
Class: short window	81	8	0.213 [0.081, 0.345]	−0.332 [−0.578, −0.086]	0.917
Class: global with exponentials	33	6	0.128 [−0.028, 0.283]	+0.269 [−0.114, 0.651]	0.498
Class: unknown (excluded from primary)	28	3	0.198 [−0.017, 0.413]	−0.244 [−0.548, 0.059]	0.971
PSR B0531+21 alone	24	1	0.604 [0.028, 1.180]	−0.740 [−1.442, −0.037]	1.138
PSR B1737−30 alone	28	1	0.279 [0.119, 0.438]	−0.727 [−0.995, −0.459]	0.910
PSR B1338−62 alone	16	1	0.133 [0.029, 0.236]	+0.006 [−0.238, 0.250]	0.401
PSR J0631+1036 alone	12	1	0.263 [−0.014, 0.539]	+0.129 [−0.494, 0.752]	0.845
D7 Crab-like (exploratory)	184	13	0.264 [0.190, 0.338]	−0.300 [−0.438, −0.161]	0.732
D7 Vela-like (exploratory)	41	8	0.331 [0.148, 0.515]	+0.294 [−0.072, 0.660]	0.499

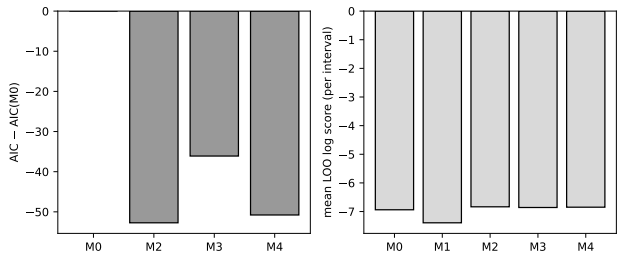


Figure 3: Model comparison. Left: AIC relative to M0. Right: mean leave-one-out log predictive score per interval.

Table 6: Post hoc single-pulsar test of $\hat{d}$: percentile of the observed value within each simulated population (1000 replicates), with the population's 95 per cent range in brackets.

	PSR B1737−30	PSR J0537−6910	PSR B0531+21
Method	SW	LR	SW
$\hat{d}$ (own M4)	−0.727	−0.182	−0.740
$\hat{\sigma}$ (own M4)	0.73	0.30	1.13
S3m	0.1 [−0.31, 0.34]	0.3 [−0.13, 0.13]	8.4 [−1.15, 1.03]
Own bias	0.0 [−0.36, 0.34]	0.5 [−0.14, 0.11]	11.3 [−1.43, 1.29]
S1m	9.5 [−1.30, 1.18]	32.6 [−0.38, 0.12]	9.8 [−1.58, 1.12]

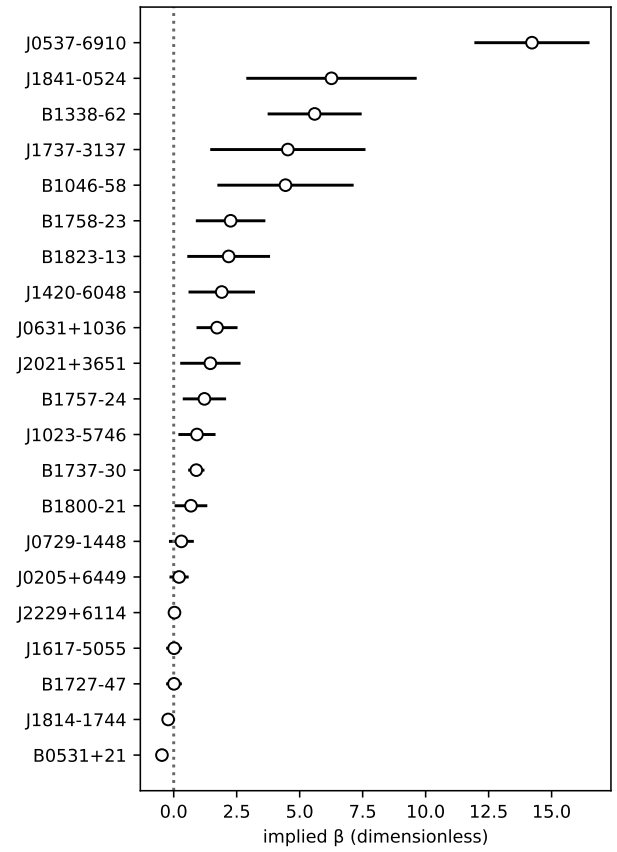


Figure 4: Implied inertia ratio β per pulsar with ± 1 standard error bars (95 per cent intervals are given in Table 4). The dotted line marks $\beta = 0$.

The result does not establish the relation quantitatively. The slope is far below unity and M1 predicts worse than intercepts alone. The emulation attributes the shortfall to attenuation by $\Delta\dot{\nu}$ measurement, but $\hat{b}$ sits at the 4.5th percentile of the emulated model-true population, and the real $\hat{\sigma} = 0.696$ is below the emulated floor of about 0.74. Both suggest that the emulation degrades $\Delta\dot{\nu}$ somewhat more than real measurements do. The verdict is relative to that emulation: a more faithful one could move the reference ranges.

9.2 Which $\Delta\dot{\nu}$?

Equation (1) refers to the step that persists after the prompt transients. The catalogue mixes that quantity with instantaneous totals (global fits) and with short-window differences that average the transients over 30–50 d (Espinoza et al., 2011).

The Crab pulsar shows how much this matters. Its catalogue $\Delta\dot{\nu}/\dot{\nu}$ values in the primary sample are a median of 16 (range 1.1–53) times the persistent step implied by $|\Delta\dot{\nu}_{\rm p}| \approx 7 \times 10^{-8}\,\Delta\nu$ (Lyne et al., 2015), because its recoveries are large ($Q \sim 0.6$–1) and fast (Wong, Backer & Lyne, 2001; Wang et al., 2012).

- Replacing the catalogue values with that persistent step moves the Crab's implied β from -0.47 to -0.04.
- Counting only the 54 per cent of the step that appears promptly (Lyne et al., 2015) gives $+0.34$.

The negative β is therefore a definitional artefact. Because a constant factor is absorbed by the per-pulsar intercept, it leaves the within-pulsar slopes untouched. This check was made after the freeze, uses an average scaling rather than per-glitch persistent steps, and is reproduced by `scripts/posthoc_crab_persistent_step.py` in the repository.

By contrast, PSR J0537−6910 is measured as a persistent step (Antonopoulou et al., 2018), matching the definition of Eq. (1). Its $\beta \approx 14$ implies, with typical $\Delta\dot{\nu}/\dot{\nu} \sim 0.5 \times 10^{-3}$, $I_{\rm B}/I \approx 1$ per cent, which is plausible for a crustal superfluid reservoir.

The two pulsars without a $\Delta\dot{\nu}$ effect may also be measurement cases, though this is an untested hypothesis. Most glitches of PSR J0631+1036 are tiny ($\Delta\nu/\nu$ of 0.4–43×10^{-9}), and three of its catalogue $\Delta\dot{\nu}/\dot{\nu}$ values are negative; a step near the short-window detection floor would attenuate d towards zero. Across its 16 primary intervals, PSR B1338−62's $\Delta\dot{\nu}/\dot{\nu}$ spans 0.20–5.00×10^{-3} while its $\Delta\nu/\nu$ spans a factor of about 700; for its seven global-fit glitches alone (Lower et al., 2021), $\Delta\dot{\nu}/\dot{\nu}$ spans only 0.22–1.06×10^{-3} over the same factor of about 700 in $\Delta\nu/\nu$. That could reflect a similar floor, or small glitches that are not creep-cycle events.

There is a tension here worth stating plainly. Zhou et al. (2022) expect equation (2) to work best for Vela-like glitches, yet the effect is clearest in two frequent, small-glitch pulsars measured by short windows, and the exploratory Vela-like group gives a positive $\hat{d}$.

9.3 The Vela pulsar

Vela would be a natural out-of-sample test, but it cannot be used with the catalogue values. For its 16 usable glitches with $\Delta\nu/\nu \geq 10^{-6}$, the catalogued $\Delta\dot{\nu}/\dot{\nu}$ ranges from about 6 to 600×10^{-3}, depending on the source:

- several values come from hours of data immediately after the glitch (e.g. the HartRAO reports of the 2010 and 2013 glitches used 14 and 9 h of data), and so are dominated by the fastest transients;
- a few, such as the four-component fit of the 2000 glitch with $\Delta\dot{\nu}/\dot{\nu} \approx 6.7 \times 10^{-3}$ (Dodson, McCulloch & Lewis, 2002), are comparable with Eq. (1).

Only one of Vela's usable glitches has an audited, admissible measurement, far below the four admissible intervals the rules require. A uniform re-measurement of Vela's persistent steps from timing data would allow a direct test.

9.4 Limitations

1. *Shared scatter.* The primary model uses one σ for all pulsars, which the single-pulsar fits contradict. The simulations share the assumption, so calibration does not correct for it. A hierarchical model with pulsar-specific scatter and partially pooled slopes would be the natural next step.
2. *Emulation.* See Section 4.3: no phase-connected fitting, no red noise, two uncited settings, and four to six pulsars, depending on σ, whose emulated uncertainties exceed the catalogue's.
3. *Sample.* PSR J0537−6910 contributes 28 per cent of the intervals, though the result survives its removal. The method behind the catalogue maintainers' own measurements is assumed (robustness case included).
4. *Constant β.* Equation (3) assumes β_p is constant within a pulsar.
5. *Self-held pre-registration.* The freeze is attested by the author's version-control history and an e-mailed record of the checksums, not by a public registry. The rules were amended sixteen times before the freeze, partly in response to simulated power.

9.5 A forward test

To obtain evidence that cannot have been tuned after the fact, I fixed on 2026 November 12, using the frozen M2 and M0 fits, forecasts of the next glitch for six pulsars. These pulsars are monitored by Jodrell Bank, have no documented end of monitoring, and have a current glitch-free stretch no longer than their longest catalogued wait. The forecasts will be scored on 2028 November 1 by the difference in log predictive density between M2 and M0 (Appendix C). With six pulsars and windows spanning years, this is a weak test, but an honest one.

10 Conclusions

A pre-registered test of the vortex-creep waiting-time relation (Zhou et al. 2022, equation 48), run once on 225 intervals in 21 pulsars and calibrated against emulations of the methods that produced the catalogue's $\Delta\dot{\nu}$, supports both its scaling and its creep-specific spin-down dependence.

The support is qualitative. The slope is strongly attenuated, the spin-down dependence varies between pulsars, and the sign of the effect is best established in PSR B1737−30 and PSR J0537−6910. The largest remaining uncertainty is the definition of $\Delta\dot{\nu}$: equation (2) needs the persistent step, while most catalogue values include transients. A uniform measurement of persistent steps would turn this qualitative test into a quantitative one.

Acknowledgements

I thank the maintainers of the Jodrell Bank Glitch Catalogue and the ATNF Pulsar Catalogue for making their data public. This work made use of NUMPY, SCIPY, PANDAS, MATPLOTLIB and LIFELINES. An AI assistant, Claude Opus 5.5 (Anthropic, 2026), used through Claude Code, contributed to the development of the analysis rules and pre-freeze amendments (D4a–D4p), wrote the analysis code under the author's instructions, reviewed computational results, checked the D4o recovery parameters against the cited literature, and assisted in drafting and revising the manuscript and in checking its statements against the analysis repository. The author directed the research, critically evaluated the work, and takes full responsibility for the paper. No AI tool is an author; the assistant's contributions are documented in the analysis repository.

Data Availability

The glitch data are public in the Jodrell Bank Glitch Catalogue. The analysis repository, `https://github.com/deeyanbothra10-droid/glitch-waiting-time-v2`, is available from the author on request. It contains the code, the catalogue snapshot used (distributed under the catalogue's own terms), the frozen rules and their checksum records, the results and figures, the forecast files, and a guide to verifying the checksums of Section 5.4.

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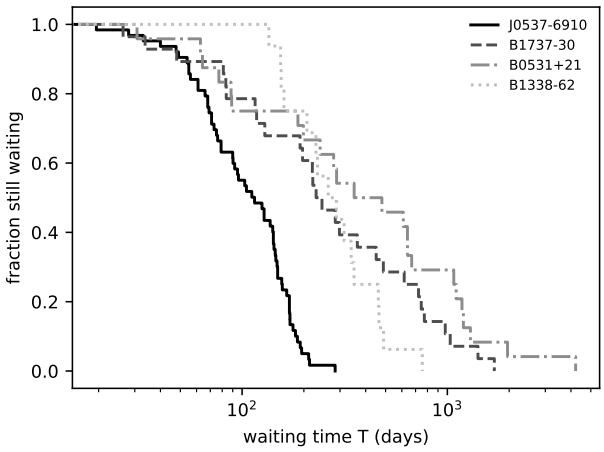


Figure 5: Kaplan–Meier estimates of the fraction of intervals still waiting, for the four pulsars with the most intervals.

A Pre-freeze amendments

Table 7 lists the amendments to the original rules (D1–D15, 2026 October 31). Every one was recorded in the decision file before the freeze; eleven state their reason explicitly, while D4b and D4d–D4g do not. None was made after a statistic relating glitch properties to real waiting times had been computed. “Prompted by simulation” marks amendments motivated by simulated power, error rates or emulator behaviour. This column is the author’s classification, based on the stated reasons where they are given, not a field recorded in the decision file.

B Source audit

C Forward forecast

The forecast was fixed on 2026 November 12 (SHA-256 of the forecast file `9f46b894...f43f0`). Two earlier drafts were superseded before any outcome could be known, because they assumed continuous monitoring for pulsars where that was not documented. They remain in the repository.

For each scored pulsar, the wait after its last catalogued glitch g is forecast as $\ln T \sim \mathcal{N}(\mu, \hat{\sigma}^2)$ from the frozen fits (M2: $\mu = c_p + \hat{b} \ln R_g$; M0: $\mu = c_p$), conditioned on no glitch before the catalogue snapshot (MJD 61343). On 2028 November 1 the catalogue will be downloaded again.

- The score is the summed log predictive density of the next glitch under M2 minus that under M0. A pulsar with no glitch by then contributes its log survival probability.
- A pulsar whose next glitch turns out to predate the snapshot is void.

Table 7: Amendments to the analysis rules before the freeze.

ID	Date	Change	Prompted by simulation
D4a	Oct 31	Permutation, Spearman and leave-one-out details	no
D4b	Oct 31	Interval conventions; simulation centring; validation cost limits; conditional-logit missingness check	no
D4c	Oct 31	Two simulation layers (S1–S3 statistics only; S1m and S4 through the emulation); emulation settings	yes (emulator behaviour)
D4d	Oct 31	S4 treated as measurement, not test; AIC rule for exponentials; decision rules calibrated against S1m and S4-mix	yes
D4e	Oct 31	Noise calibration to formal uncertainties; replicate inclusion rule	yes
D4f	Oct 31	Short-window class created from the source audit, with a fixed admission rule	no (audit)
D4g	Oct 31	Documented censoring dates; RXTE/NICER gap of PSR J0537−6910; Jodrell Bank robustness case	no (audit)
D4h	Oct 31	Intercepts corrected for covariate spread; false-positive bound tied to $\alpha = 0.025$	yes
D4i	Oct 31	Tiered outcomes; implied β made report-only	yes (power)
D4j	Oct 31	Exclusive outcome order; artefact outcome redefined so that a null result is not labelled an artefact	yes
D4k	Oct 31	Failed-replicate rule; σ grid extended to 0.8 and 1.2	yes
D4l	Nov 1	Directional, simulation-calibrated T1 and T2; corrected bounds for C and A	yes (error rates)
D4m	Nov 1	Measured size-only null S3m; A1 relabelled as not specific to creep; calibrated intervals	yes
D4n	Nov 1	Generating σ matched to fitted scatter; T2 based on $\hat{d}$ against null references	yes (power)
D4o	Nov 1	Literature-based recovery parameters replace placeholders	no (literature)
D4p	Nov 1	Search bounds, robustness reporting, outcome labels, removal of provisional tags	no

Table 8: Audited sources of admissible $\Delta\dot{\nu}$ values. N is the glitch count recorded for each source in the project's source-audit file (`data/meta/source_audit.csv`); it is a bookkeeping count from the audit worklist, not a count of catalogue entries, and its derivation was not recorded. "Our measurement" denotes the catalogue maintainers' own values, whose method is assumed to follow Basu et al. (2022).

Source	N	Telescope	Cadence (d)	Class	$\Delta\dot{\nu}$ as reported
Espinoza et al. (2011)	103	Lovell	2–10	short window	difference of pre- and post-glitch solutions at the epoch; post-glitch span ideally 30–50 d; transients not separated
Wang et al. (2000)	15	Parkes	14–42	short window	pre- and post-glitch solutions over ∼100–300 d, extrapolated to the epoch
Antonopoulou et al. (2018)	45	RXTE	<14	local relaxation	persistent step; fits bounded by neighbouring glitches
Ho et al. (2026)	23	NICER	variable	local relaxation	glitch model fitted per segment; recovery terms where detected
Yu et al. (2013)	17	Parkes	14–28	local relaxation	segment fits centred on the glitch; recoveries where detected
Basu et al. (2022)	36	Lovell	10–14	global	$\Delta\dot{\nu}_\mathrm{p} + \sum \Delta\nu_\mathrm{d}/\tau_\mathrm{d}$; all glitches fitted simultaneously
Lower et al. (2021)	20	Parkes	∼30	global	glitch and red-noise parameters fitted together; recoveries free
Gügercinoğlu et al. (2022)	8	*Fermi*-LAT	1–30	global	spin and glitch parameters fitted together, with exponentials
Our measurement	17	Lovell	10–14	global (assumed)	method not published

Table 9: Scored forecasts: median and central 50 per cent window of the next-glitch date under M2 and M0.

Pulsar	Model	Median	50% window
PSR B0531+21	M2	2027 May 26	2027 Jan 23 – 2028 Jan 04
	M0	2027 Jul 18	2027 Feb 12 – 2028 May 05
PSR J2229+6114	M2	2028 Jul 31	2027 Sep 13 – 2030 Jan 08
	M0	2028 Jun 04	2027 Jul 23 – 2029 Dec 09
PSR J1737−3137	M2	2028 Dec 08	2027 Sep 11 – 2031 Mar 08
	M0	2029 Apr 20	2027 Oct 30 – 2032 Jan 23
PSR J2021+3651	M2	2029 Oct 04	2028 Feb 08 – 2032 Aug 06
	M0	2029 Aug 09	2027 Dec 25 – 2032 Jul 28
PSR B1800−21	M2	2030 Nov 23	2028 Jul 08 – 2035 Feb 19
	M0	2030 Jun 21	2028 Apr 18 – 2034 Jul 20
PSR B1823−13	M2	2031 May 17	2028 Sep 22 – 2036 Jan 26
	M0	2032 Apr 24	2029 Jan 29 – 2038 Apr 13